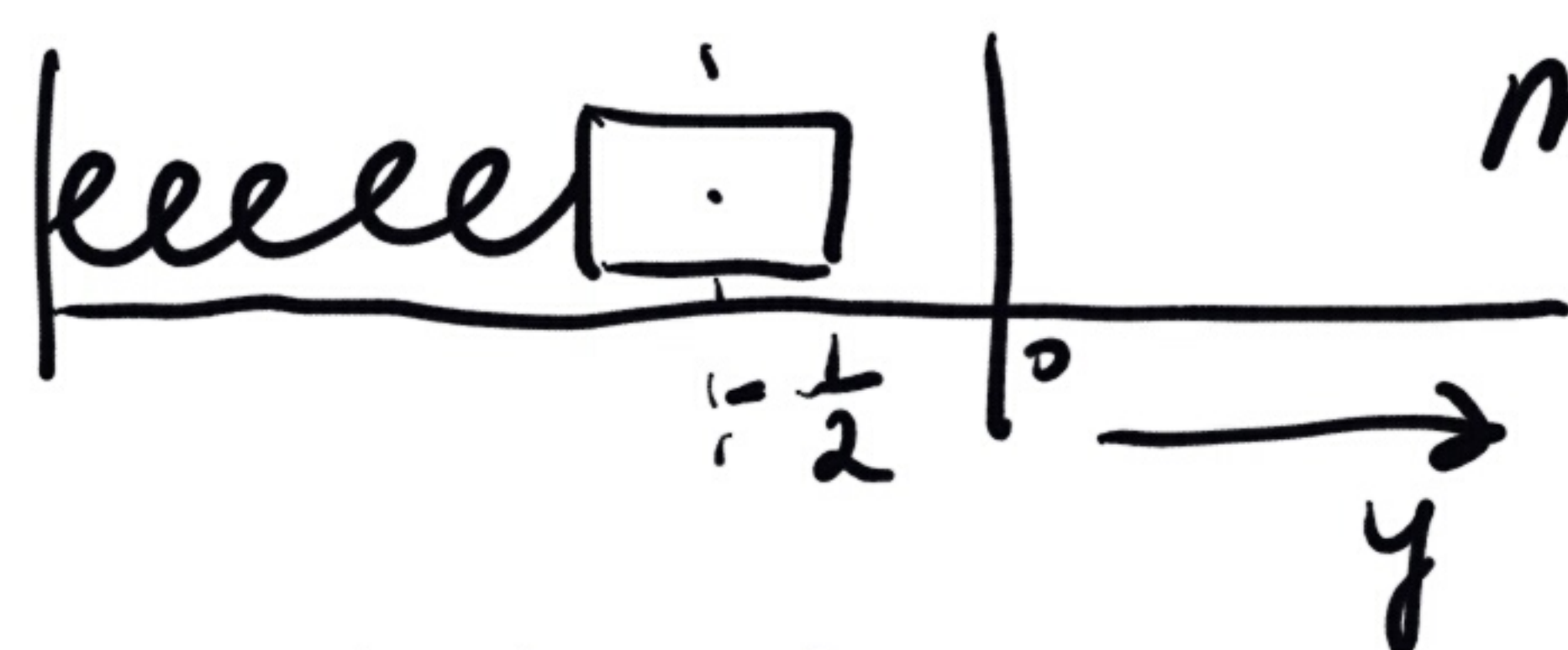


Example 3 A $1/4$ -kg mass is attached to a spring with a stiffness 4 N/m as shown in Figure 4.31(a). The damping constant b for the system is 1 N-sec/m. If the mass is displaced $1/2$ m to the left and given an initial velocity of 1 m/sec to the left, find the equation of motion. What is the maximum displacement that the mass will attain?

(p 221)



$$m = \frac{1}{4}, k = 4, b = 1, y(0) = -\frac{1}{2}$$

$$y'(0) = -1$$

$$y(t) = ?$$

$$\frac{1}{4}y'' + y' + 4y = 0 \Rightarrow y'' + 4y' + 16y = 0$$

$$y_{\max} = ?$$

$$r^2 + 4r + 16 = 0, r_{1,2} = \frac{-4 \pm \sqrt{16 - 4 \cdot 16}}{2} \Rightarrow$$

$$r_{1,2} = -2 \pm 2\sqrt{3}i$$

$$y(t) = e^{-2t} (C_1 \cos 2\sqrt{3}t + C_2 \sin 2\sqrt{3}t)$$

$$y'(t) = -2e^{-2t} (C_1 \cos 2\sqrt{3}t + C_2 \sin 2\sqrt{3}t) + e^{-2t} (-2\sqrt{3}C_1 \sin 2\sqrt{3}t + 2\sqrt{3}C_2 \cos 2\sqrt{3}t)$$

$$y(0) = -\frac{1}{2} \Rightarrow -\frac{1}{2} = C_1, y'(0) = -1 \Rightarrow -1 = -2C_1 + 2\sqrt{3}C_2$$

$$\Rightarrow 2\sqrt{3}C_2 = -1 + 2C_1 = -1 - 1 = -2 \Rightarrow C_2 = -\frac{1}{\sqrt{3}}$$

$$y(t) = e^{-2t} \left(-\frac{1}{2} \cos 2\sqrt{3}t - \frac{1}{\sqrt{3}} \sin 2\sqrt{3}t \right)$$

$$y'(t) = -2e^{-2t} \left(-\frac{1}{2} \cos 2\sqrt{3}t - \frac{1}{\sqrt{3}} \sin 2\sqrt{3}t \right) + e^{-2t} \left(\sqrt{3} \sin 2\sqrt{3}t - 2 \cos 2\sqrt{3}t \right) = 0 \quad (\cdot e^{2t})$$

$$\cos 2\sqrt{3}t + \frac{2}{\sqrt{3}} \sin 2\sqrt{3}t + \sqrt{3} \sin 2\sqrt{3}t - 2 \cos 2\sqrt{3}t = 0$$

$$-\cos 2\sqrt{3}t + \left(\frac{2}{\sqrt{3}} + \sqrt{3} \right) \sin 2\sqrt{3}t = 0$$

$$\frac{5}{\sqrt{3}} \sin 2\sqrt{3}t = \cos 2\sqrt{3}t$$

$$\Rightarrow \tan 2\sqrt{3}t = \frac{\sqrt{3}}{5} \Rightarrow 2\sqrt{3}t = \arctan \frac{\sqrt{3}}{5}$$

$$\Rightarrow y_{\max} = y(t_0)$$

$$t_0 = \frac{\arctan \frac{\sqrt{3}}{5}}{2\sqrt{3}}$$